

PROBLEM STATEMENT #50 | Developer Tools & IT Operations

Centralized Audit Trail Compliance Engine

1. Problem Context & Overview

A compliance logging platform that ingests enterprise application event logs, masks sensitive PII fields in accordance with GDPR, and generates cryptographically verifiable immutable audit exports.

Target Stakeholders / Actors: *Compliance Officer, Security Auditor*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall parse structured JSON audit logs, identify and mask PII fields (Social Security Numbers, Credit Cards) before persisting records to storage.

Sample Acceptance Criteria: Pass: Masked audit log saved with SHA-256 integrity hash; Fail: Cleartext credit card number stored in database.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: Audit log query endpoints must support complex filtering across 10 million historical records in under 1 second.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.